

Option-maximizing co-constitution

Option-maximizing co-constitution: experience, observation points, and the unity of observer and cosmos

Author: Akito Shiraishi **Draft:** PhilArchive philosophy version, EN v0.1 (2026-06-03). Companion physics version on arXiv (gr-qc). **Stance:** A first-person experiential account, scaffolded on Tegmark, Wheeler, and Wigner, advanced to an explicit metaphysical thesis (OMCC). Evaluated by unifying economy (pragmatic structural realism), not by truth-correspondence—so the unfalsifiability objection is structurally disarmed, while in-frame predictions remain testable.

Abstract

Why does the mathematics an observer discovers coincide with the mathematics of the cosmos? Wigner called this coincidence inexplicable. Tegmark answers that the cosmos *is* a mathematical structure and the observer a self-aware substructure (so coincidence is necessary, with the cosmos prior); Wheeler answers that the observer constitutes the cosmos (participatory, so coincidence is authored, with the observer prior). These conflict. We propose **Option-Maximizing Co-constitution (OMCC)**: observer and cosmos are neither prior, but **co-constituted within a self-exciting cycle that runs toward maximizing the option quantity Ω (future reachable states)**; their mathematics coincide because they occupy the same phase of one cycle. The infinite regress “why the Ω -direction?” is closed not as a linear wall (the Past Hypothesis) but as a **cycle** (Penrose’s conformal cyclic cosmology; autopoiesis; Wheeler’s self-exciting circuit). The thesis’s distinctive content—beyond the three named figures—is the **observation point** (the finest observing unit, recurring fractally across micro and macro), the **Ω -maximization drive**, the **solenoid geometry**, and a **first-person experiential datum** (a directly seen nested cycle) that plays the philosophical role of adjudicating the Tegmark–Wheeler tension.

1. The problem (Wigner) and two answers in tension

Wigner (1960): the effectiveness of mathematics in describing nature borders on the mysterious, with no rational explanation. Two later positions answer, but conflict: - **Tegmark (MUH)**: external reality *is* a mathematical structure; observers are self-aware substructures (SAS). Coincidence is necessary—**cosmos prior**. - **Wheeler**

(**participatory universe; “It from Bit”; self-exciting circuit**): observer-participancy constitutes reality; the universe evolves observers through which it observes itself into being. Coincidence is authored—**observer prior**.

The tension is the chicken-and-egg of priority.

2. OMCC: the co-constitution cycle

Observer and cosmos are neither prior. They are **co-constituted within a self-exciting cycle running toward Ω -maximization**. The observer’s mathematics coincides with the cosmos’s because both are the same cycle viewed at the same phase (synchrony). Wigner’s mystery dissolves once description (observer-mathematics) and the described (cosmos-mathematics) are seen as one cycle’s self-reference.

This sublates Tegmark (realism) and Wheeler (participation) as two phases of one cycle—akin to Whitehead’s mutual immanence, but with the **drive specified as Ω -maximization** and the **geometry specified as a solenoid**, which is the author’s own contribution.

3. Observation points (a contribution absent in the three figures)

The **observation point** is the finest observing unit from a given vantage; an individual is a weighted set of observation points with energy allocations; and observation points **recur fractally across micro and macro scales**. This refines Tegmark’s SAS by giving it (i) a scale-recursive structure and (ii) an internal energy/attention allocation. On a torus, an observation point is a point whose dense (irrational-winding) trajectory scans the whole—formalizing the experiential “stage-4” pervasive access.

4. Experience as primary datum (the experiential method)

In a non-ordinary state the author directly *saw* a “nested cycle”—the image that motivates OMCC. The methodological claim: rather than decoration, **this experience does philosophical work**—because the cycle was seen, it can serve as the candidate structure that resolves the Tegmark–Wheeler tension. The scaffolding (Tegmark/Wheeler/Wigner) is used as authority/premise; the experience is the first-person datum advanced on top. We separate two senses of “fact”: the experience is a **phenomenological fact** (certain qua experience); its identity with cosmic structure is a **hypothesis** (partly testable via cosmic topology, in the companion physics version). The brain-side fact (the neural correlates of the experience) and the cosmos-side fact are distinct; OMCC predicts their coincidence because the observing brain is co-constituted within the cosmic cycle.

5. Unity = truth (the thesis's epistemic core)

The mathematical structure a self-aware observation point discovers coincides with the mathematical structure of the cosmos that contains it; **this coincidence is the core of truth.**

Tegmark holds coincidence *necessary*; the author advances one step—**coincidence as the definition of truth's core**. This is a stance, not a proof; it answers Wigner by fiat (declaring the synchrony point to be where truth lives), which is a philosophical choice evaluated by fruitfulness, not correspondence.

6. Closing the regress as a cycle

“Why the Ω -direction?” regresses, in the physics companion, to the Past Hypothesis (a wall). Philosophically, OMCC closes it as a **cycle**: Penrose's conformal cyclic cosmology (an aeon's high- Ω far future conformally supplies the next low-entropy beginning), **autopoiesis** (self-production's closed loop), and Wheeler's **self-exciting circuit** all say the regress is not a line ending at a wall but a self-supplying loop. The ultimate “why is there a cycle at all?” remains (the metaphysical wall is not removed, only relocated), and we say so.

7. Honest epistemic status (pragmatic structural realism)

OMCC is **metaphysics evaluated by unifying economy, not truth-correspondence**. This disarms the unfalsifiability objection at its premise (the objection assumes a theory must correspond and be refutable to have value; we evaluate by how economically the frame unifies). But we keep a strict duality: the **frame** is judged by economy, while **in-frame empirical claims** (the companion's cosmic-topology hook; the *refuted* solenoid spectral-dimension prediction) remain testable and were tested—one was refuted and is reported as such. We explicitly refuse to claim universal truth (“this is the author's frame, not a universal doctrine”), forestalling the cultic failure mode (no external verifier). The named-authority coincidence (independently reaching Tegmark/Wheeler/Wigner/Penrose) attests to the quality of the intuition; the missing piece was descriptive language, here supplied.

8. Conclusion

OMCC offers a single organizing idea—co-constitution of observer and cosmos within an Ω -maximizing cycle—that sublates the Tegmark–Wheeler tension, answers Wigner by locating truth at the synchrony point, closes the regress as a cycle, and contributes the observation point, the Ω -drive, the solenoid geometry, and a first-person experiential datum. It is metaphysics, openly: evaluated by unifying economy, anchored to one falsifiable observational hook (cosmic topology, companion paper) and one honestly refuted prediction. We publish it for external verification and extension.

References

Wigner (1960) *Comm. Pure Appl. Math.* 13:1; Tegmark (2008) *Found. Phys.*, (2014) *Our Mathematical Universe*; Wheeler (1990) “Information, Physics, Quantum”; Penrose (2010) *Cycles of Time*; Maturana & Varela (1972) *Autopoiesis and Cognition*; Whitehead, *Process and Reality*; Husserl (co-constitution); Worrall (structural realism); James/Peirce (pragmatism); companion physics version (arXiv).